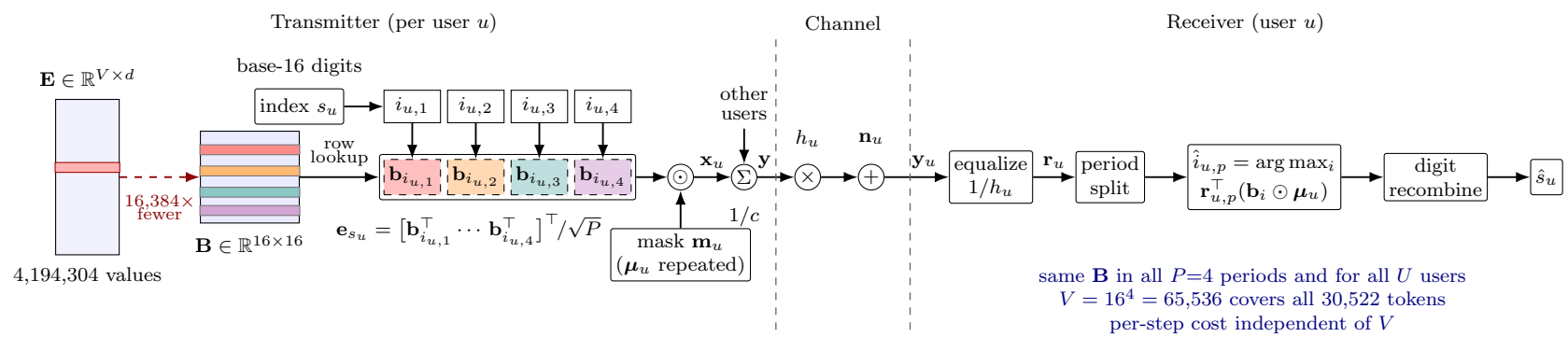

Transmitter (per user u)
Channel
Receiver (user u)
$\mathbf{E} \in \mathbb{R}^{V \times d}$
4,194,304 values
16,384× fewer
$\mathbf{B} \in \mathbb{R}^{16 \times 16}$
base-16 digits
index s_u
$i_{u,1}$ $i_{u,2}$ $i_{u,3}$ $i_{u,4}$
row lookup
$\mathbf{b}_{i_{u,1}}$ $\mathbf{b}_{i_{u,2}}$ $\mathbf{b}_{i_{u,3}}$ $\mathbf{b}_{i_{u,4}}$
$\mathbf{e}_{s_u} = [\mathbf{b}_{i_{u,1}}^\top \cdots \mathbf{b}_{i_{u,4}}^\top]^\top / \sqrt{P}$
mask $\mathbf{m}_u$ ($\boldsymbol{\mu}_u$ repeated)
$\mathbf{x}_u$
other users
$1/c$
$\mathbf{y}$
h_u
$\mathbf{n}_u$
$\mathbf{y}_u$
equalize $1/h_u$
$\mathbf{r}_u$
period split
$\hat{i}_{u,p} = \arg\max_i \mathbf{r}_{u,p}^\top (\mathbf{b}_i \odot \boldsymbol{\mu}_u)$
digit recombine
$\hat{s}_u$
same $\mathbf{B}$ in all P=4 periods and for all U users
$V = 16^4 = 65{,}536$ covers all 30,522 tokens
per-step cost independent of V